

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 204 — Past Questions & Materials
by Renance

Course: MTH 204
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 200

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Paper 1 of 1 — MTH 204 LINEAR ALGEBRA 1 2023,2024

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN.

COLLEGE OF SCIENCE :

DEPARTMENT OF MATHEMATICS

FIRST SEMESTER EXAMINATION 2023/2024 SESSION

COURSE TITLE & CODE: LINEAR ALGEBRA | (MTH 213)

Instruction: Attempt any four questions Time: 2 hours

Question 1 .

25 34 : f -3 -2

Given the following matrices $A = \begin{bmatrix} 3 & 1 \\ 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 6 & 2 \\ 1 & 4 \end{bmatrix}$, find $A + B$.

4 3 ho oh A (atten Paar

$$ae \cdot fj = z$$

Evaluate

(a) (i) (BA)? (ii) (BC)? (9 marks)

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(b) Reduce the matrix $B = 4a : '$ to its echelon and hence find its rank. (8.5marks)

$$5 \ 8 \ 1 = 2$$

Question 2

Bu 5) 7-2

Find the determinant of the matrix A if $\det(A) = 5$.

(a) Find the determinant of the matrix A. (10 marks)

Bg ae

(b) List 5 properties of determinants and justify each property with an example. (7.5 marks)

Question 3

$$i^2 = i;$$

(a) Find the inverse of A (A) given that $A = \begin{pmatrix} 4 & 5 \\ 12 & 5 \end{pmatrix}$ (12.5 marks)

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(b) Given $B = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, find the trace of the transpose of the matrix BC . (5 marks)

$$LQ = a$$

Question 4

(a) Let $X = \mathbb{R}^n$ and $A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$. Define $T: X \rightarrow X$ by $Tz = Az$, where $z = \begin{pmatrix} x \\ y \end{pmatrix} \in X$. Show that T is a linear transformation and hence find the Kernel and Range of T (12.5 marks)

(b) Let $A = (2, 3, 4)$ and $B = (8, -2, 5)$ be two vectors, find the cross (vector) product of A and B ($A \times B$)

(5 mars)

Question 5

(a) Define the following (i) Natural num!

ctor W as a linearly combination of the vectors V1, V2 and V3.

bers (ii) Subspace (iii) Integers (iv) Rational numbers (4 marks)

(b) Write the ve (13.5 marks)

$Y = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $m = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$, $B = \begin{pmatrix} 3 \\ 7 \end{pmatrix}$ endl $7 \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ Wa | |

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Question 6

(a) Define a vector space and list all its axioms. (6 marks)

and V3 are linearly independent, where (11.5 marks) -

(b) Determine if the vectors VV2

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